Real-Time Temperature Prediction Technique of Aluminium Electrolysis Furnace Based on the KSI Principle

[Author names and affiliations to be inserted]
Corresponding author: [email]

Abstract

In this paper, we propose a new real-time fuzzy prediction modeling method based on the moving rate of uncertainty reasoning (the KSI principle) and apply it to the real-time temperature prediction of aluminium electrolysis furnaces. Conventional regression-based adaptive identification cannot adequately represent the strong nonlinearity of the electrolysis process, while modern deep-learning predictors require large training data sets, expensive retraining, and offer limited interpretability to furnace operators. Unlike Zadeh's max–min compositional rule of inference, the proposed method constructs a Takagi–Sugeno (T–S) fuzzy model whose rule activation is computed from moving rates of s-type, z-type, and trapezoidal membership functions, and identifies the conclusion parameters by a recursive least-squares procedure that is consistent with the moving-rate-based product reasoning. The method extends our previous works published in the *Iranian Journal of Fuzzy Systems* (16(3), pp. 17–34, 2019), *WSEAS Transactions on Computer Research* (8, pp. 73–89, 2020), and the *Romanian Journal of Information Science and Technology* (24(3), pp. 257–283, 2021) from the single-input single-output (SISO) case to a multi-input single-output (MISO) real-time modeling framework that requires only one thermocouple sensor. The technique was implemented in MATLAB, MySQL, and C++ and verified through simulation and field experiments on 60 industrial aluminium electrolysis furnaces. The prediction accuracy of the proposed model exceeds 99% (e.g., 99.86% with a mean error of 0.315 °C for furnace No. 2), whereas an ANFIS model of the same furnace attains 99.76% with a mean error of 2.289 °C using five times more rules, and earlier patented predictors reach only about 85%. The experimental results show that the new modeling method is considerably more effective than the previous ones.

Keywords: fuzzy reasoning; temperature prediction; prediction model; moving rate; KSI principle; T–S fuzzy model; aluminium electrolysis furnace

1 Introduction

The bath temperature of an aluminium electrolysis furnace directly reflects the energy balance of the cell and is one of the most important process variables for stable and energy-efficient aluminium production: it governs current efficiency, ledge formation, anode effects, and the service life of the cell lining. Keeping the electrolyte temperature within its narrow optimal operating window is therefore critical, yet the temperature is conventionally measured by thermocouple only at long intervals (about every three hours in practice), because continuous immersion measurement in the corrosive cryolite melt is impractical and costly. A reliable *real-time prediction* of the furnace temperature from easily measured electrical signals (cell voltage and line current) is consequently of great industrial value.

An adaptive identification algorithm based on a regressive model was introduced to solve the real-time temperature prediction problem [1]. However, such a regressive model, which represents the characteristic of the aluminium electrolysis furnace with a single analytic expression, is not appropriate for representing the strong nonlinearity of the object and has low accuracy. To solve this problem, many works have been devoted to representing the nonlinearity of objects effectively, ranging from computational-fluid-dynamics simulation [4] and real-time thermal prediction in enclosures [3] to data-driven machine learning [10, 11, 12, 13, 14, 15, 16, 17, 18] and fuzzy reasoning and modeling [2, 5, 6, 7, 19, 20, 21, 22, 23, 24, 25, 26, 29, 30].

In [4], a new fuzzy modeling method based on superalloy heating in a resistance electrolysis furnace was introduced and the effectiveness of the method was verified. However, it is impossible to use the fuzzy modeling method of [4] for the estimation of conclusion parameters. Otherwise, unlike [4], in [5] a new fuzzy reasoning method based on a compensating operation, the so-called moving rate, was proposed and applied to fuzzy neural network systems. Based on [5], in [6] the authors presented a fuzzy reasoning method based on a distance measure and discussed its reductive property, and in [7] a new fuzzy modus ponens and modus tollens based on moving distance in the SISO fuzzy system was proposed.

In [8, 9], a fuzzy identification algorithm for temperature prediction of the aluminium electrolysis furnace was proposed and a prediction was made for the temperature of the aluminium electrolysis furnace for three hours. The accuracy of the prediction is not high, and the approach is expensive because it uses three sensors for one aluminium electrolysis furnace.

In this paper, we present a new method for real-time temperature prediction of aluminium electrolysis furnaces by applying fuzzy reasoning based on the moving rate of uncertainty reasoning proposed in [5, 6, 7] (which we call the *KSI principle*) to the fuzzy model identification problem. In other words, our method is based on the moving rate of uncertainty reasoning. We then compare our method with previous methods through simulation experiments and real tests in the field.

The main contributions of this paper are summarized as follows: (i) moving rates are defined for s-type, z-type, and trapezoidal membership functions, so that the rule-activation strength of a T–S fuzzy model can be computed directly from the input information without the max–min composition; (ii) a four-stage recursive identification procedure of the conclusion parameters is derived that is fully consistent with the moving-rate-based product reasoning, thereby removing the contradiction contained in the fourth estimation stage of the classical product-reasoning identification of [1]; (iii) the SISO moving-rate reasoning of [5, 6, 7] is extended to a MISO dynamic T–S model suitable for real-time application with a single thermocouple sensor per furnace; and (iv) the method is verified on 60 industrial furnaces against ANFIS and the patented predictors [8, 9], achieving model accuracy above 99% with a mean deviation two to three times smaller than that of ANFIS.

The remainder of this paper is organized as follows. Section 2 reviews related work on temperature prediction of aluminium electrolysis furnaces and on fuzzy approximate reasoning, and states the research gap. Section 3 describes the new fuzzy modeling method based on the moving rate. Section 4 presents the temperature prediction model of the aluminium electrolysis furnace and its experimental verification. Section 5 discusses the results, and Section 6 concludes the paper.

2 Related Work and Motivation

Research on modeling and prediction for the aluminium electrolysis process has been active for five decades; a recent bibliometric analysis of the field from 1970 to 2023 documents a marked acceleration of intelligent, data-driven studies in the last few years [16]. Existing approaches can be grouped into three families: physics-based numerical models, data-driven machine-learning models, and knowledge-based fuzzy models.

2.1 Physics-based and numerical modeling

First-principles approaches describe the thermal state of the cell by heat- and mass-balance equations. Moazamigoodarzi *et al.* developed real-time temperature predictions in IT server enclosures using reduced thermal models [3], showing that physically grounded models can run in real time when the geometry is simple. For high-temperature metallurgical objects, Fu *et al.* proposed a new modeling method for superalloy heating in a resistance electrolysis furnace using FLUENT-based computational fluid dynamics [4]. Such models provide valuable physical insight, but for an industrial aluminium electrolysis cell they require detailed knowledge of geometry, material properties, and boundary conditions, demand heavy computation that conflicts with real-time use, and cannot easily adapt online when the cell condition drifts (ledge profile change, anode replacement, lining ageing). Moreover, as noted above, the conclusion-parameter estimation required for adaptive prediction cannot be carried out within the framework of [4].

2.2 Data-driven machine-learning approaches

With the accumulation of production data, machine learning has been widely applied to the aluminium electrolysis process. Zhou *et al.* predicted anode effects with a singular-value-thresholding and extreme-gradient-boosting approach [10]; Lu *et al.* predicted the liquidus temperature of complex electrolyte systems by machine-learning methods [11]; Cao *et al.* proposed an online prediction method for the molten aluminium height based on a kernel extreme learning machine [12]; and Chen *et al.* carried out time-series prediction of iron and silicon content in the metal [13]. For temperature itself, Lei *et al.* designed a self-supervised deep LSTM network for industrial temperature prediction in aluminium processes [14], and Zhu *et al.* combined a dual-attention LSTM for the non-stationary sub-sequence with an ARMA model for the stationary sub-sequences to predict the cell temperature, alleviating the low accuracy and hysteresis of earlier soft sensors [15]. Lundby *et al.* modeled the coupled mass and energy balance of the cell with sparse neural networks with skip connections, which outperformed plain multilayer perceptrons especially for long prediction horizons and limited data [18]. Most recently, Lin *et al.* predicted the bath temperature of a 500-kA reduction cell with five machine-learning models and found that LightGBM performs best, reaching an average difference of 1.6 °C between predicted and measured temperature, with SHAP analysis used to interpret the model [17].

These studies confirm the strong nonlinearity of the temperature dynamics and the feasibility of data-driven prediction. Nevertheless, deep and ensemble models share several practical drawbacks for furnace-side deployment: they need large historical data sets and periodic offline retraining; their many parameters make online recursive adaptation difficult; their black-box character limits acceptance by operators (which is precisely why post-hoc explanation tools such as SHAP had to be added in [17]); and per-cell customization of a potline with hundreds of cells is costly. This motivates transparent rule-based models whose parameters can be updated recursively in real time.

2.3 Fuzzy approximate reasoning and fuzzy modeling

Fuzzy set theory offers exactly such transparency. Since Zadeh introduced the concept of the linguistic variable and its application to approximate reasoning [1], the compositional rule of inference (CRI) and the T–S fuzzy model have become standard tools of fuzzy modeling and fuzzy control [2]. However, CRI-type reasoning does not, in general, satisfy the reductive property, and the classical T–S identification based on product reasoning contains a contradiction in its fourth estimation stage [1, 6], so that many refinements of the inference mechanism have been studied.

One active line of research replaces composition by distance- or similarity-based inference. Luo and Wang studied interval-valued fuzzy reasoning full-implication algorithms based on t-representable t-norms [19]; Duan and Li analysed similarity measures of intuitionistic fuzzy sets and their applications [21]; and Liu and Li proposed quintuple intuitionistic fuzzy implication reasoning algorithms [22]. Fuzzy rule interpolation, which infers conclusions for sparse rule bases from the distances between fuzzy sets, has also progressed rapidly: Li *et al.* demonstrated interpretable mammographic mass classification with fuzzy interpolative reasoning [20], and Xu *et al.* advanced dynamic fuzzy rule interpolation via density-based spatial clustering of interpolated outcomes [23]. These recent studies confirm that distance-driven inference remains a central topic of approximate reasoning.

Our group has developed a coherent alternative stream based on the *moving rate* (compensating operation). A fuzzy reasoning method based on the compensating operation and its application to fuzzy systems was proposed in [5]; a fuzzy reasoning method based on a distance measure with a proven reductive property was presented in [6]; fuzzy modus ponens and modus tollens based on moving distance in the SISO fuzzy system were established in [7]; an extended-distance-measure reasoning method for discrete fuzzy set vectors of different dimensions was reported in [24]; and an approximate reasoning method based on the least common multiple with its property analysis was given in [25]. Collectively we refer to this reasoning principle—rule activation computed from the moving rate of the input with respect to the antecedent fuzzy sets—as the *KSI principle*. The principle has recently been extended to interval fuzzy modeling and applied to turf color image evaluation [26], to precipitation forecasting and disaster warning [28], and to agricultural parameter selection [27]. In parallel, explainability and tuning of fuzzy systems continue to develop, e.g., through causal-entropic Bayesian frameworks for robust fuzzy inference [29] and improved shark smell optimization with adaptive parameter scheduling for fuzzy controller tuning [30].

For the aluminium electrolysis furnace in particular, fuzzy identification predictors were patented in [8, 9], but their model accuracy is only about 85% and three temperature sensors per furnace are required, while the interval-measurement updating scheme of [2] did not reach 91% accuracy.

2.4 Research gap and contributions

The above review reveals a clear gap. Physics-based models [3, 4] are accurate in principle but computationally heavy and not adaptive online. Machine-learning models [10, 11, 12, 13, 14, 15, 17, 18] achieve good accuracy but require large data sets and lack the transparency and cheap recursive adaptation demanded by potline operation. Classical fuzzy identification [1, 2] is transparent but suffers from the contradiction of its estimation stage and from low accuracy when a single global expression is used, and the existing furnace-specific fuzzy predictors [8, 9] are both inaccurate and sensor-expensive. Finally, our previous moving-rate reasoning methods

[5, 6, 7, 24, 25] were formulated for the SISO case and had not yet been connected to recursive T–S parameter identification for a real industrial MISO object. The present paper closes this gap by building a moving-rate-based (KSI-principle) T–S modeling method with consistent recursive least-squares identification and by validating it on 60 industrial aluminium electrolysis furnaces with a single thermocouple sensor per furnace.

3 A New Fuzzy Modeling Method Based on the Moving Rate

Unlike Zadeh's max–min composition principle [1], the new fuzzy modeling method based on the moving rate [5, 6, 7] is as follows.

The moving rates for s-type, z-type, and trapezoidal membership functions are defined as follows. For input information x_{j0}, the moving rates d_{ij} of rule i and input j are defined by formulas (1), (2), and (3), respectively.

For the s-type membership function with breakpoints $x_{aij} < x_{cij}$,

$$d_{ij} = \begin{cases} \dfrac{x_{cij} - x_{j0}}{x_{cij} - x_{aij}}, & \text{if } x_{aij} \le x_{j0} < x_{cij}, \\ 0, & \text{if } x_{j0} \ge x_{cij}, \\ 1, & \text{if } x_{j0} < x_{aij}. \end{cases} \tag{1}$$

For the z-type membership function with breakpoints $x_{aij} < x_{cij}$,

$$d_{ij} = \begin{cases} 0, & \text{if } x_{j0} \le x_{aij}, \\ \dfrac{x_{j0} - x_{aij}}{x_{cij} - x_{aij}}, & \text{if } x_{aij} < x_{j0} \le x_{cij}, \\ 1, & \text{if } x_{j0} > x_{cij}. \end{cases} \tag{2}$$

For the trapezoidal membership function with breakpoints $x_{1ij} < x_{2ij} < x_{3ij} < x_{4ij}$,

$$d_{ij} = \begin{cases} 1, & \text{if } x_{j0} < x_{1ij} \text{ or } x_{j0} \ge x_{4ij}, \\ \dfrac{x_{2ij} - x_{j0}}{x_{2ij} - x_{1ij}}, & \text{if } x_{1ij} \le x_{j0} \le x_{2ij}, \\ 0, & \text{if } x_{2ij} \le x_{j0} \le x_{3ij}, \\ \dfrac{x_{j0} - x_{3ij}}{x_{4ij} - x_{3ij}}, & \text{if } x_{3ij} \le x_{j0} \le x_{4ij}. \end{cases} \tag{3}$$

The recursive matrix construction procedure of the T–S fuzzy model, which is based on the product reasoning, is as follows.

Stage 1. For input information x_{j0}, get the moving rates d_{ij} using formulas (1)–(3).

Stage 2. Get the product term d_i of each rule using the following formula (4):

$$d_i = 1 - \left[d_{i1} \wedge d_{i2} \wedge \cdots \wedge d_{im}\right], \qquad i = \overline{1,n},\ j = \overline{1,m}. \tag{4}$$

Stage 3. Calculate the rate $\hat{d}^i$ of each rule affecting the output using the following formula

(5):

$$\hat{d}^i = \frac{d_i}{\sum\limits_{i=1}^{n} d_i}. \tag{5}$$

Stage 4. The fuzzy model output y^0, which is calculated by formula (6), determines the recursive matrix Z using formula (7):

$$y^0 = \sum_{i=1}^{n} \hat{d}^i y^i = \sum_{i=1}^{n} \bigl(a_0^i z_0^i + a_1^i z_1^i + \cdots + a_m^i z_m^i\bigr), \tag{6}$$

$$Z = \begin{pmatrix} z_0^{1(1)} & \cdots & z_m^{1(1)} & z_0^{2(1)} & \cdots & z_m^{2(1)} & \cdots & z_0^{n(1)} & \cdots & z_m^{n(1)} \\ z_0^{1(2)} & \cdots & z_m^{1(2)} & z_0^{2(2)} & \cdots & z_m^{2(2)} & \cdots & z_0^{n(2)} & \cdots & z_m^{n(2)} \\ \vdots & & \vdots & \vdots & & \vdots & & \vdots & & \vdots \\ z_0^{1(p)} & \cdots & z_m^{1(p)} & z_0^{2(p)} & \cdots & z_m^{2(p)} & \cdots & z_0^{n(p)} & \cdots & z_m^{n(p)} \end{pmatrix}. \tag{7}$$

The fuzzy modeling method based on the moving rates proceeds T–S fuzzy modeling with input–output data as follows. Denote the coefficients of formula (6) with a vector

$$A = \bigl(a_0^1\ a_1^1\ \cdots\ a_m^1\ a_0^2\ \cdots\ a_m^n\bigr)^T. \tag{8}$$

The T–S fuzzy model formula (9) is denoted with formula (10) simply:

$$R^i:\ \text{if } x_1 \text{ is } A_1^i,\ x_2 \text{ is } A_2^i,\ \ldots,\ x_m \text{ is } A_m^i\ \text{ then }\ y^i = a_0^i + \sum_{j=1}^{m} a_j^i x_j, \tag{9}$$

where R^i $(i = \overline{1,n})$ is the ith object rule, $x^0 = [x_1^0, x_2^0, \ldots, x_m^0]^T$ is the input vector, A_j^i is the jth antecedent fuzzy set of the ith rule, and y^i is the conclusion value of R^i. By fuzzy reasoning, the set $y = [y_1, y_2, \ldots, y_p]^T$ of p real output values may be expressed as formula (10):

$$y = ZA + \varepsilon. \tag{10}$$

Here, p is the number of input–output data couples, $z_j^{i(k)}$ $(i = \overline{1,n},\ j = \overline{0,m},\ k = \overline{1,p})$ is the kth z_j^i value, and $\varepsilon = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_p)^T$ is the corresponding error vector, with

$$\begin{cases} z_0^{i(k)} = \hat{d}^i, \\ z_j^{i(k)} = \hat{d}^i x_j^{0(k)}, \end{cases} \qquad i = \overline{1,n},\ j = \overline{1,m},\ k = \overline{1,p}. \tag{11}$$

Here, $x_j^{0(k)}$ is the jth element of the kth input vector, n is the number of rules, and m is the dimension of the input information. Construct the recursive matrix by using the fourth stage of product reasoning. The estimation formulas proposed in reference [1] contradict each other, and it is impossible to use its conclusion-parameter estimation. Because θ_k in the formula (12) is a row vector and $F_k(y_k - H_k\theta_{k-1})$ is a column vector, matrix addition is impossible there. The corrected recursive estimation is

$$\theta_k = \theta_{k-1} + F_k\bigl(y_k - H_k\theta_{k-1}\bigr), \tag{12}$$

$$F_k = \frac{S_{k-1}H_k^T}{1 + H_k S_{k-1} H_k^T}, \tag{13}$$

$$S_k = S_{k-1} - F_k H_k S_{k-1}. \tag{14}$$

Here θ_k is the parameter vector to estimate and S_k is the standard deviation matrix; H_k is the data vector defined as below in formula (15):

$$H_k = \left(W_k^1,\, W_k^1 x_{1k}, \ldots, W_k^1 x_{mk},\; W_k^2,\, W_k^2 x_{1k}, \ldots, W_k^2 x_{mk},\; \ldots,\; W_k^n, \ldots, W_k^n x_{mk}\right). \tag{15}$$

The results that investigate the conclusion-parameter estimation formula of reference [1] and obtain the new estimation formula are as follows. For the dynamic T–S fuzzy system represented as formula (16), the conclusion-parameter vector is estimated by formula (17):

$$\begin{aligned}
R^i : \text{IF } & y(k-1) \text{ is } A_1^i,\ y(k-2) \text{ is } A_2^i,\ \ldots,\ y(k-n_1) \text{ is } A_{n_1}^i, \\
& u_1(k-t_{d1}) \text{ is } A_{n_1+1}^i,\ \ldots,\ u_1(k-t_{d1}-n_1) \text{ is } A_{n_1+n_1+1}^i,\ \ldots, \\
& u_p(k-t_{dp}) \text{ is } A_{n_1+\cdots+n_{p-1}+p}^i,\ \ldots,\ u_p(k-t_{dp}-n_p) \text{ is } A_{n_1+\cdots+n_p+p}^i, \\
\text{THEN } & y^i(k) = a_0^i + a_1^i y(k-1) + a_2^i y(k-2) + \cdots + a_{n_1}^i y(k-n_1) \\
& \qquad + a_{n_1+1}^i u_1(k-t_{d1}) + \cdots + a_{n_1+n_1+1}^i u_1(k-t_{d1}-n_1) + \cdots \\
& \qquad + a_{n_1+\cdots+n_p+p}^i\, u_p(k-t_{dp}-n_p),
\end{aligned} \tag{16}$$

$$\theta_k = \theta_{k-1} + \left(y_k - H_k \theta_{k-1}^T\right)^T \frac{S_{k-1} H_k^T}{1 + H_k S_{k-1} H_k^T}, \qquad S_k = S_{k-1} - F_k H_k S_{k-1}, \quad F_k = \frac{S_{k-1} H_k^T}{1 + H_k S_{k-1} H_k^T}. \tag{17}$$

Here R^i is the ith fuzzy rule, A_j^i is the jth fuzzy subset of R^i, y^i is the conclusion reasoning output value, a_j^i is a conclusion parameter, $u_1(\cdot), \ldots, u_p(\cdot)$ are input variables, $y(\cdot)$ is the output variable, k is the time variable, and $t_{d1}, \ldots, t_{dp}$ are the input time delays.

4 Temperature Prediction Model of the Aluminium Electrolysis Furnace and Its Verification

For making the temperature prediction model of the aluminium electrolysis furnace, we proceeded in the following stages.

Stage 1. Measure the temperature of the aluminium electrolysis furnace by thermocouple and construct the database set of input and output by extracting the voltage and current from the current data server.

Stage 2. Applying the proposed fuzzy modeling method to this database set, make the temperature prediction model and evaluate its correctness by comparison with the previous fuzzy modeling method.

Stage 3. Evaluate whether the obtained temperature prediction model represents the temperature characteristic of the aluminium electrolysis furnaces correctly or not, and verify its validity.

The fuzzy modeling program for making the temperature prediction model is programmed by MATLAB, and the real-time temperature prediction model is realized by C++. By applying the dynamic T–S fuzzy modeling method based on the moving-rate product reasoning, we made the temperature prediction models of 60 aluminium electrolysis furnaces. The output value of the temperature prediction model and the real value for aluminium electrolysis furnace No. 2 by the proposed method is shown in Figure 1.

In Figure 1, the accuracy of the model is 99.86% and the mean error between the model

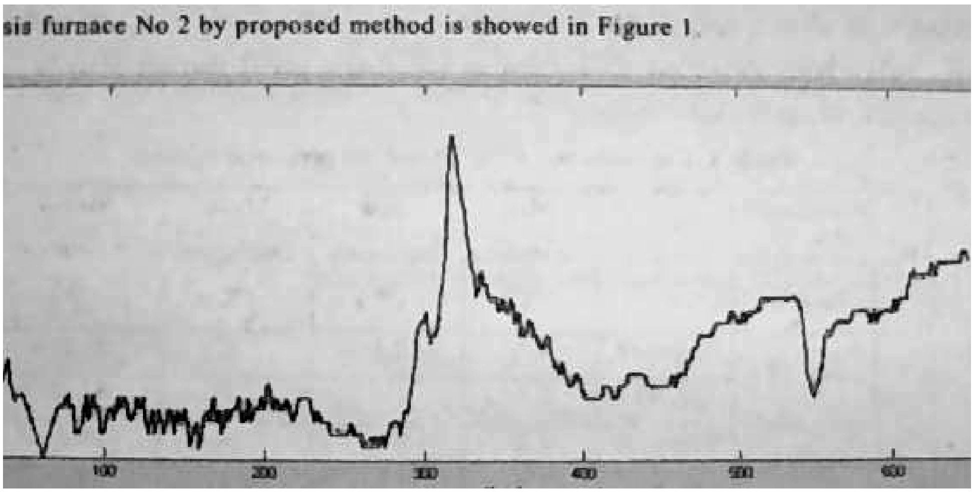

Figure 1: The output value of the temperature prediction model and real value for aluminium electrolysis furnace No. 2 (proposed method).

output and the real value is 0.315 °C. Figure 2 shows the output value of the fuzzy model obtained by ANFIS and the real output value in aluminium electrolysis furnace No. 2. The accuracy of the model of Figure 2 is 99.76% and the mean error between the output value and the real value is 2.289 °C.

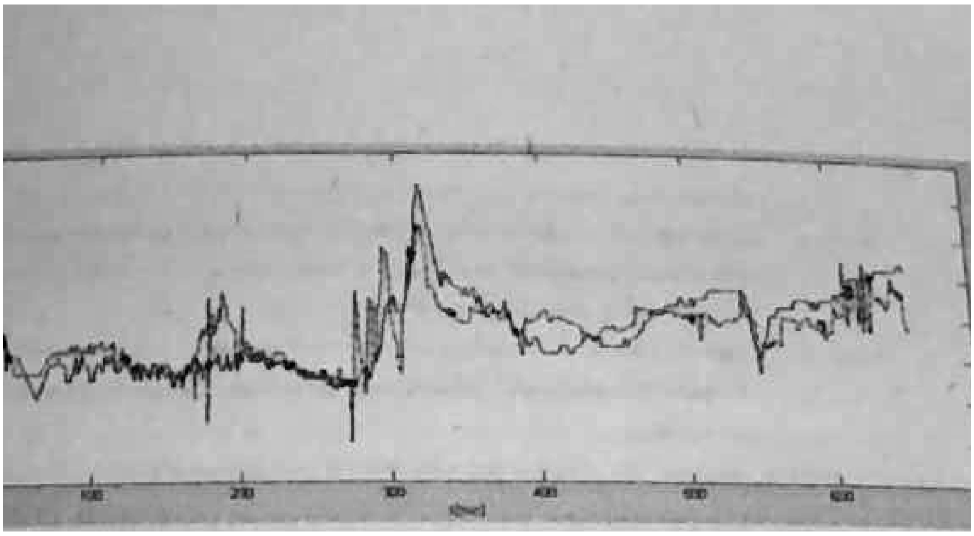

Figure 2: Accuracy of the model obtained by ANFIS for aluminium electrolysis furnace No. 2.

The number of rules of the model obtained by ANFIS is 25, but by the proposed fuzzy modeling method it is only 5, and the mean error of our proposed method is as small as $2 \sim 3$ times that of ANFIS in the simulation experiment result, so we can conclude that applying the proposed method is very efficient. Table 1 shows the comparison of the experimental results that we obtained in the field with the previous result [8, 9] and the proposed method.

5 Discussion

Table 1 shows the comparison between the previous result and the proposed method. Considering the table, the model accuracy of the previous method [8, 9] is about 85%, while that of the

Table 1: Comparisons of [8, 9] and the proposed method.

No	Aluminium Electrolysis Furnace	Method	Max Deviation (°C)	Min Deviation (°C)	Mean Deviation (°C)	Model Accuracy (%)
1	No. 15	Proposed	4.6	0.1	1.57	99.84
		[8, 9]	4.8	1.1	2.8	88.45
2	No. 48	Proposed	6.2	0.8	2.94	99.71
		[8, 9]	7.3	1.6	2.4	83.67
3	No. 89	Proposed	11.1	0.4	6.05	99.40
		[8, 9]	8.6	1.9	2.6	82.56

proposed method is over 99%. Moreover, although the previous method [2] proceeds with model updating by interval measurement, the accuracy of the model did not reach 91%. The cause is that previous research tries to represent the aluminium electrolysis furnace model—a strongly nonlinear object—by only one analytic formula, as mentioned above.

Compared with recent machine-learning predictors, the proposed model is also competitive: the LightGBM model of [17] attains an average difference of 1.6 °C on a 500-kA cell using rich production features, while our five-rule fuzzy model reaches a mean error of 0.315 °C for furnace No. 2 using only voltage and current signals, and its rule base remains directly interpretable by operators without post-hoc explanation tools. In addition, the recursive estimation (12)–(14) updates the conclusion parameters sample by sample, so no offline retraining is required when the cell state drifts, unlike the deep models of [14, 15, 18]. Overall, through the simulation experiments in MATLAB and the field experiments realized by matching MATLAB, MySQL, and C++, it was verified practically that the application of the proposed method proceeded successfully.

6 Conclusion

In this paper, a new real-time fuzzy prediction modeling method based on the moving rate of uncertainty reasoning (the KSI principle) was proposed and applied to the real-time temperature prediction of aluminium electrolysis furnaces. Moving rates were defined for s-type, z-type, and trapezoidal membership functions; a four-stage recursive matrix construction procedure of the T–S fuzzy model was derived; the contradiction in the conclusion-parameter estimation of the classical product-reasoning identification was corrected; and a dynamic MISO T–S model driven by voltage and current signals with a single thermocouple sensor was built. Verification on 60 industrial aluminium electrolysis furnaces showed model accuracy above 99% with a mean deviation two to three times smaller than that of ANFIS with five times fewer rules, and far above the roughly 85% accuracy of the previously patented fuzzy predictors.

In future work, we will extend the proposed method to interval and type-2 fuzzy models following the interval extension of the KSI principle [26], combine it with metaheuristic parameter scheduling such as improved shark smell optimization [30] for automatic tuning of the membership breakpoints, and integrate the prediction model into model-predictive temperature control of the potline.

References

[1] Zadeh, L. A.: The concept of a linguistic variable and its applications to approximate reasoning, I, II, III. *Information Sciences* 8, pp. 199–249, 301–357; 9, pp. 43–80, 1975.

[2] Zhang, H., et al.: *Fuzzy Modeling and Fuzzy Control*, pp. 33–62, 2009.

[3] Moazamigoodarzi, H., Pal, S., Ghosh, S., Puri, I. K.: Real-time temperature predictions in IT server enclosures. *International Journal of Heat and Mass Transfer* 127, pp. 890–900, 2018.

[4] Fu, Z., Yu, X., Shang, H., Weng, Z., Zhang, Z.: A new modeling method for superalloy heating in resistance electrolysis furnace using FLUENT. *International Journal of Heat and Mass Transfer* 128, pp. 679–687, 2019.

[5] Kwak, S.-I., et al.: A fuzzy reasoning method based on compensating operation and its application to fuzzy systems. *Iranian Journal of Fuzzy Systems* 16(3), pp. 17–34, 2019. DOI: 10.22111/IJFS.2018.4175.

[6] Kwak, S.-I., Kim, U.-H., Kim, K.-J., Son, I.-M.: Fuzzy reasoning method based on distance measure and its reductive property. *WSEAS Transactions on Computer Research* 8, pp. 73–89, 2020. DOI: 10.37394/232018.2020.8.10.

[7] Kwak, C.-J., Ri, K.-C., Kwak, S.-I., Kim, K.-J., Ryu, U.-S., Kwon, O.-C., Kim, N.-H.: Fuzzy modus ponens and tollens based on moving distance in SISO fuzzy system. *Romanian Journal of Information Science and Technology* 24(3), pp. 257–283, 2021.

[8] Li, J.: Fault predictor for aluminium electrolysis. Chinese Invention Patent Publication Specification, pp. 1–8, 2006 (in Chinese).

[9] Luo, Y.: On-site measuring device for the primary-crystal (liquidus) temperature of aluminium electrolyte. Chinese Utility Model Patent Specification, pp. 1–12, 2010 (in Chinese).

[10] Zhou, K., Zhang, Z., Liu, J., Hu, Z., Duan, X., Xu, Q.: Anode effect prediction based on a singular value thresholding and extreme gradient boosting approach. *Measurement Science and Technology* 30(1), pp. 1–11, 2019. DOI: 10.1088/1361-6501/aaee5e.

[11] Lu, H., Hu, X., Cao, B., Chai, W., Yan, F.: Prediction of liquidus temperature for complex electrolyte systems Na_3AlF_6-AlF_3-CaF_2-MgF_2-Al_2O_3-KF-LiF based on the machine learning methods. *Chemometrics and Intelligent Laboratory Systems*, pp. 110–120, 2019. DOI: 10.1016/j.chemolab.2019.03.015.

[12] Cao, B., Cui, J., Li, Q., Wang, M., Li, X., Yan, Q.: Online prediction method of molten aluminium height in electrolytic cell based on extreme learning machine with kernel function. *Mathematical Problems in Engineering* 2021, 9980194, 2021. DOI: 10.1155/2021/9980194.

[13] Chen, L., Wu, Y., Liu, Y., Liu, T., Sheng, X.: Time-series prediction of iron and silicon content in aluminium electrolysis based on machine learning. *IEEE Access* 9, pp. 10699–10710, 2021. DOI: 10.1109/ACCESS.2021.3050548.

[14] Lei, Y., Karimi, H. R., Chen, X.: A novel self-supervised deep LSTM network for industrial temperature prediction in aluminum processes application. *Neurocomputing* 502, pp. 177–185, 2022. DOI: 10.1016/j.neucom.2022.06.080.

[15] Zhu, X., et al.: Temperature prediction of aluminum reduction cell based on integration of dual attention LSTM for non-stationary sub-sequence and ARMA for stationary subsequences. *Control Engineering Practice*, 2023.

[16] Li, X., Lin, J., Liu, C., Liu, A., Shi, Z., Wang, Z., Jiang, S., Wang, G., Liu, F.: Research on aluminum electrolysis from 1970 to 2023: a bibliometric analysis. *JOM* 76(7), pp. 3265–3274, 2024. DOI: 10.1007/s11837-024-06596-1.

[17] Lin, J., Liu, A., Wang, Z., Shi, Z., Liu, F.: Bath temperature prediction of aluminum reduction cell based on machine learning algorithm. *Journal of Sustainable Metallurgy* 11, pp. 1419–1430, 2025. DOI: 10.1007/s40831-025-01027-0.

[18] Lundby, E. T. B., et al.: Sparse neural networks with skip-connections for modeling the mass and energy balance of aluminum electrolysis cells, 2023.

[19] Luo, M., Wang, Y.: Interval-valued fuzzy reasoning full implication algorithms based on the t-representable t-norm. *International Journal of Approximate Reasoning* 122, pp. 1–8, 2020. DOI: 10.1016/j.ijar.2020.03.009.

[20] Li, F., Shang, C., Li, Y., Shen, Q.: Interpretable mammographic mass classification with fuzzy interpolative reasoning. *Knowledge-Based Systems* 191, 105279, 2020. DOI: 10.1016/j.knosys.2019.105279.

[21] Duan, J., Li, X.: Similarity of intuitionistic fuzzy sets and its applications. *International Journal of Approximate Reasoning* 137, pp. 166–180, 2021.

[22] Liu, Z., Li, X.: Quintuple intuitionistic fuzzy implications reasoning algorithms and application. *Iranian Journal of Fuzzy Systems* 20(1), pp. 153–170, 2023.

[23] Xu, R., Lin, J., Shang, C., Shen, Q.: Towards dynamic fuzzy rule interpolation via density-based spatial clustering of interpolated outcomes. *Proceedings of the IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2023.

[24] Son, I. M., Kwak, S. I., Han, U. J., Pak, J. H., Han, M., Pyon, J. R., Ryu, U. S.: A novel fuzzy approximate reasoning method based on extended distance measure in SISO fuzzy system. arXiv:2003.13450, 2020.

[25] Kwak, S.-I., et al.: New approximate reasoning method based on least common multiple and its property analysis. *Proceedings of the 5th International Conference on Applied Science, Engineering and Management*, 2022.

[26] Kong, H.-U., Kwak, S.-I., et al.: New interval fuzzy modeling method based on KSI principle and its application in turf color image evaluation. *Journal of Industrial and Management Optimization*, pp. 1–23, 2025. DOI: 10.3934/jimo.2023146.

[27] Kim, U.-H., Kwak, S.-I., Jang, U.-J.: Investigation and selection of growth parameters of rice using FCM algorithm. *Information Technologies: Problems and Solutions*, 2025.

[28] Kim, I.-G., Kwak, S.-I.: Implementation and result analysis of precipitation forecast and disaster warning system. *Information Technologies: Problems and Solutions*, 2025.

[29] Choe, J.-H., Jang, Y.-J., Kwak, S.-I., Ri, O.-S., Kong, H.-U.: Causal-entropic fuzzy inference: a Bayesian framework for explainable and robust fuzzy reasoning. *Journal of Fuzzy Systems and Control* 4(1), 2026. ISSN 2986-6537. DOI: 10.59247/jfsc.v4i1.392.

[30] Choe, J.-H., Choe, G.-C., Kwak, S.-I., Chae, H.-G., Kong, H.-U.: Improved shark smell optimization with adaptive parameter scheduling for fuzzy controller tuning. *Academy Journal of Science and Engineering* 20(1), 2026.